

1. Prove that if $A \neq B$, then $A \times B \neq B \times A$.

Set up so sets $A \neq B$.

$$A = \{1, 2, 3\}, B = \{1, 2, 4\}$$

Cartesian product, so each element is ordered pair.

$$A \times B = \{ (1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (3, 1), (3, 2), (3, 4) \}$$

$$B \times A = \{ (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (4, 1), (4, 2), (4, 3) \}$$

As highlighted above, $A \times B \neq B \times A$ as some ordered pairs are different in both cartesian products. Additionally, order matters significantly in cartesian products.

2. Prove that $(A \cup B) - C = (A - C) \cup (B - C)$ using iff and logical equivalence.

$x \in (A \cup B) - C$	—
iff $(x \in A \vee x \in B) - C$	Definition of union
iff $(x \in A \vee x \in B) \wedge x \notin C$	Definition of difference
iff $(x \in A \vee x \in B) \wedge \neg(x \in C)$	Definition of $\notin$
iff $(x \in A \wedge \neg(x \in C)) \vee (x \in B \wedge \neg(x \in C))$	Distributive law of logic
iff $(x \in A \wedge \neg(x \in C)) \cup (x \in B \wedge \neg(x \in C))$	Definition of union
iff $(x \in A \cap \neg(x \in C)) \cup (x \in B \cap \neg(x \in C))$	Definition of intersection
iff $(x \in A \cap x \notin C) \cup (x \in B \cap x \notin C)$	Definition of $\notin$
iff $(x \in A \cap x \notin C) \cup (x \in B \cap x \notin C)$	Definition of set difference
iff $x \in (A - C) \cup (B - C)$	—

Since $x \in (A \cup B) - C$ iff $x \in (A - C) \cup (B - C)$, therefore $(A \cup B) - C = (A - C) \cup (B - C)$.

3A. Disprove: If $A \cup C \subseteq B \cup C$ then $A \subseteq B$.

Proving by counterexample, I'll first set up sets representing A, B, and C.

$$A = \{4\}, B = \{1, 2\}, C = \{4\}$$

Now I abide by rules of union and combine the sets.

$$A \cup C = \{4, 4\}, B \cup C = \{1, 2, 4\}$$

Then the subset can be evaluated! It is a proper subset.

$$\{4, 4\} \subseteq \{1, 2, 4\} \rightarrow A \cup C \subseteq B \cup C$$

Now that the first condition is met, let's check the second one.

$$\{4\} \not\subseteq \{1, 2\}$$

A is not a subset of B as 4 is not an element of B. Therefore by counterexample, the statement: $A \cup C \subseteq B \cup C$ then $A \subseteq B$ must be false.

3B. If $A \cap C \subseteq B \cap C$ then $A \subseteq B$.

Disproving by counterexample, I'll first set up the sets representing A, B, and C.

$$A = \{1, 2, 3\}, B = \{1, 2, 5\}, C = \{1, 2\}$$

Now I will abide by the rules of intersection and combine the sets.

$$A \cap C = \{1, 2\}, B \cap C = \{1, 2\}$$

Thus the subset can be evaluated as true as the sets are identical and contain the same elements.

$$\{1, 2\} \subseteq \{1, 2\} \rightarrow A \cap C \subseteq B \cap C$$

Now let's check the second condition.

$$\{1, 2, 3\} \not\subseteq \{1, 2, 5\} \mid A \not\subseteq B$$

A is not a subset of B as 3 is not an element of B. Therefore by counterexample,

If $A \cap C \subseteq B \cap C$ then $A \subseteq B$ must be false.

4. Prove that if $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$ then $A \subseteq B$.

- Assume that $A \not\subseteq B$ for proof of contradiction and $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$ is all true.
- Because of this, there exists some element $x \in A$ and $x \notin B$.
- If $A \cup C \subseteq B \cup C$ is true, then x must also be in C as every element of $A \cup C$ must be in $B \cup C$ as that is definition of subset.
- If $A \cap C \subseteq B \cap C$ is also true, then x must also be in B . By definition of intersection, x must be in set B as x has already been established to be in the intersection of A and C .
- This is a **CONTRADICTION** as it contradicts the previous statement $A \not\subseteq B$ and $x \notin B$. Therefore, the assumption is false and if $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$, then $A \subseteq B$ must be true.

5. $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$ using subset argument.

Case I: $x \in (A \cup B) - (A \cap B)$. Prove that $x \in (A - B) \cup (B - A)$

- By definition of set difference and union, I can substitute and simplify the expression.

$$(x \in A \text{ or } x \in B) \text{ and } (x \notin A \cap B)$$

- By definition of intersection, I can further simplify the expression into two conditions, where x either belongs to A and not B or B and not A .

$$x \in A \text{ and } x \notin B \text{ or } x \in B \text{ and } x \notin A$$

- Through definition of set difference, I can reinsert the symbol to show that x belongs to A and not B (vice versa)

$$x \in (A - B) \text{ or } x \in (B - A)$$

- By definition of union, I can substitute the union for the or.
 $x \in (A - B) \cup (B - A)$
- Identical to other condition, therefore is a subset.
 $(A \cup B) - (A \cap B) \subseteq (A - B) \cup (B - A)$

Case II: $x \in (A - B) \cup (B - A)$. Prove that $x \in (A - B) \cup (B - A)$.

- Definition of set difference and definition of union.
 $(x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$
 x belongs to A or B but not both, so $x \notin A \cap B$
 Rewrite as $x \in (A \cup B) - (A \cap B)$ through definitions of union, set difference, and intersection.
 Proves that $(A - B) \cup (B - A) \subseteq (A \cup B) - (A \cap B)$.

6. Prove that $f(n) = 5n + 9$ is one-to-one, where the domain and co-domain of f is $\mathbb{Z}^+$. Show that f is not onto.

One to One:

- If a function is one-to-one, then there will be different outputs for every input in the function.
- Let m and n be positive integers in the domain of the function ($\mathbb{Z}^+$)
- Assume that $f(m) = f(n)$ if the function is injective, so that $5n + 9 = 5m + 9$
- Simplify into $m = n$, which shows that every input's output will be different unless two inputs are the same.

Onto:

- In order to prove that it is not onto, I must find an element in the codomain ($\mathbb{Z}^+$) that cannot be set equal to $5n + 9$ and return a positive integer for n (domain).
- Take 2 for example. Set $2 = 5n + 9$ and it returns $n = -7/5$
- $-7/5$ is not a positive integer, therefore there is at least one positive integer element in the co-domain that cannot be represented by the domain of the function.
- Therefore, the function is not onto.

7. Prove that $f(m, n) = m + n + mn$ is onto, where the domain of f is $\mathbb{Z} \times \mathbb{Z}$ and the co-domain of f is $\mathbb{Z}$. Show that f is not one-to-one.

Onto:

- If a function is onto, then there is a value in the co-domain for each value in the domain of the function.
- If the function is onto, then the range will equal the co-domain.
- Range = $\{3, 5, 8, 11, 15, 19, 24, 29, 35\}$ Co-Domain = $\mathbb{Z}$
- $m(n + 1) = y - n \rightarrow m = (y - n) / (n + 1) \mid n = (y - m) / (1 + m)$
- Function f is onto as any pair will be satisfied through an appropriate codomain value, as evidenced by the individual equations for m and n .

Not one to one:

- If a function is one to one, then every input has a different output.
- Since the domain is cartesian, all of the inputs come in coordinate-like pairs, just like there was (m, n) in the problem.
- Abiding by this, allow for a test for $f(m, n) = f(n, m)$, switching the x and y values.
- $m + n + mn = n + m + nm \rightarrow mn = nm$. This will equate to the same value.
- Since $f(m, n)$ and $f(n, m)$ or any coordinate pair that is switched around will equate to the same value in the co-domain, this function is certainly not one to one.